CS/PRJ/565 — PROJECT TIMELINE

Global Impact and Information Dissemination Strategy Assessment Tool for SETI Post-Detection Communication
CS4099 Senior Honours Project · University of St Andrews

Semester 1 exams 5 – 21 Dec 2026 **Semester 2 exams** 8 – 22 May 2027 **Feature freeze** mid-March 2027 **Module deadlines** add from handbook

1 SETUP

Mid-September – early October 2026

[ ] DOER form submitted and signed
[ ] AI declaration confirmed with supervisor + Projects Coordinator
[ ] Reading: GDELT Event Codebook 2.0 + CAMEO codebook
[ ] Reading: draft Declaration update (arXiv:2510.14506)
[ ] Reading: Rio 2.0
[ ] Reading: Nardo et al., *Constructing Composite Indicators*
[ ] Reading: V-Dem codebook
[ ] Source audit complete, tier assigned per source
[ ] Repo scaffolded, CI green
[ ] First real GDELT pull (--limit 8, then a full day)
[ ] Country-code crosswalk working, with tests

→ **DOER in; ingestion slice running on live data**

2 INGESTION & REFERENCE LAYER

October 2026

[ ] Indicator framework table drafted (dimension / source / variable / tier / direction / justification)
[ ] Double-counting check: which sources sit inside WGI's 35?
[ ] Scheduled ingestion (cron or APScheduler), retry and dedup
[ ] Raw and processed storage layers settled
[ ] V-Dem and WGI downloaded, versions in VERSIONS.md
[ ] Reference table joins to country-day data on ISO3
[ ] Archive accumulating daily — this is the fall-back dataset

→ **Data accumulating; joined country table produced**

3 SCORING

November 2026

[ ] Normalisation implemented (min-max or z-score, justified)
[ ] Weighting and aggregation — consider geometric aggregation: stops one strong dimension masking a collapsed one
[ ] Fast/slow split: structural baseline + current deviation
[ ] Uncertainty bands from V-Dem and WGI standard errors
[ ] Unit tests on the scoring maths
[ ] Run provenance (run ID, config, data versions, scores)

→ **Scores for all countries, with uncertainty**

4 CONSOLIDATE

1 – 4 December 2026

[ ] Stabilise, document, tag a release
[ ] End-of-semester demo
[ ] NO NEW FEATURES until January

— EXAMS AND BREAK

5 December 2026 – mid-January 2027

[ ] Draft background chapter while the reading is fresh
[ ] Draft methodology chapter — the indicator framework table goes in near-verbatim
[ ] No coding. Protect the exam period.

5 AI LAYER

Semester 2, weeks 1–3

[ ] Classification approach chosen and justified
[ ] Hand-annotated gold set (a few hundred items)
[ ] Agreement measured (kappa or Krippendorff's alpha)
[ ] Outputs cached and deterministic — must be reproducible
[ ] Fast-signal component feeding into readiness scores

→ **AI layer live, with measured accuracy**

6 FRONTEND

Semester 2, weeks 4–6

[ ] Map with choropleth readiness scores
[ ] Expandable per-country indicator breakdown
[ ] Time slider / assessment date selection
[ ] Configurable weights + "Run" button — recompute over stored data only; if it makes a network call, the architecture has gone wrong
[ ] Score explanation: which dimensions drove this result
[ ] Admin panel for reference refresh (secondary objective)

→ **Usable by someone who did not write it**

7 EVALUATION

Semester 2, weeks 7–8

[ ] Historical replays: Venus phosphine, BLC1, Oumuamua, plus a non-SETI case (e.g. early COVID-origin debate)
[ ] State predictions in advance, then compare with what happened
[ ] Sensitivity analysis (SALib): which assumptions drive output?
[ ] Article-weighting sensitivity: weighted vs unweighted
[ ] FEATURE FREEZE — mid-March 2027

→ **Replays + sensitivity carry the evaluation — make both systematic**

8 WRITE-UP

Mid-March – April 2027

[ ] Bug fixes only — no new features
[ ] Dissertation complete
[ ] Evaluation chapter
[ ] Submission

WHY THIS ORDER

The engine precedes the interface because the interface depends on it. The frontend comes last because it is the component most likely to expand without limit. Freezing in mid-March protects writing time, which is where most projects get squeezed. The throwaway map in Phase 1 still matters: it forces interface decisions into the open while the data model can still change cheaply.

STANDING HABITS

- Dated project log in the repo, a few lines per session.
- Archive every raw pull — by February that is months of your own frozen corpus, exactly the fall-back the DOER promises.
- Record every methodological decision as you make it.

OPEN DECISIONS TO RESOLVE

- Article weighting: weighted vs unweighted tone
- Unlocated events: currently dropped — check what fraction that is
- V-Dem: latest-year-per-country (mixes vintages) vs fixed year
- Severity parameter: condition readiness on announcement magnitude? Would make the Rio scale genuinely load-bearing in the design

ETHICS

No approval required (agreed with supervisor). Keep the aggregates-and-URLs storage design — it is what keeps that position defensible.